

- (r_t) and state (s_t) change.
3. Update the reward (r_t) and state (s_t) values.
4. Calculate the next action using the Q-learning formula.
5. Update Q values.

Among the three learning approaches, supervised learning is used only when both the input and output data is available along with the data model. Unsupervised learning is used only when the input data is available, and reinforcement learning is required when there is no training data and the agent is required to be trained from the working environment. Apart from these, there are many other learning approaches like self-learning, feature learning, sparse dictionary learning, anomaly detection, robot learning, and association rule learning.

Hardware and software for ML

To run any computer program (algorithm) a central processing unit (CPU) is required. Most of the CPU hardware contains multiple cores, but such CPUs are designed for serial operations and don't provide high throughput. However, a graphics processing unit (GPU) can have a higher number of cores compared to a CPU and has higher throughput also. A CPU has more cache memory that

can be used for complex operations, but a GPU can be used for simple operations even if it has lesser cache memory. That's why a GPU can be used for with the help of compute unified device architecture (CUDA), which is also known as General Purpose computing on Graphics Processing Units (GPGPU).

Even though a GPU is useful for ML, it has certain limitations, one of which is that its architecture cannot be customised for specific purposes. To overcome such limitations, a special kind of chip (hardware) is used, called the field programmable gate array (FPGA). Such a chip can be programmed as per the purpose, and its architecture can be customised for the specific agent in ML. An FPGA chip can contain thousands of memory units, which are more than in a GPU and can give better throughputs. Another advantage of using an FPGA for ML is hardware acceleration, which can accelerate certain parts of an algorithm, making it more efficient than a GPU.

In order to take full advantage of the hardware, good ML software can help build ML models as per the requirement. There is a lot of such software available online – both proprietary and free. Among the free software, TensorFlow, Shogun, Apache Mahout, PyTorch,

KNIME and Keras are more widely used and are the popular ones.

- TensorFlow can help build ML solutions through its extensive interface of a CUDA GPU. It provides support and functions for various applications of ML such as computer vision, NLP and reinforcement learning. This software is best suited for beginners in ML, and is also used for education purposes too.
- Shogun is free software that supports languages like Python, R, Scala, C#, Ruby, etc. It supports vector agents, dimensionality reduction, clustering algorithms, hidden Markov models and linear discriminant analysis.
- Apache Mahout is popular software that provides expressive Scala DSL and a distributed linear algebra framework for deep learning computations and native solvers for CPUs, GPUs as well as CUDA accelerators.
- PyTorch was developed by Facebook's AI Research lab (FAIR) and is mainly used for ML applications such as computer vision and natural language processing. It provides Tensor computing (like NumPy) with strong acceleration via GPUs, and supports deep neural networks built on a tape based automatic differentiation system.

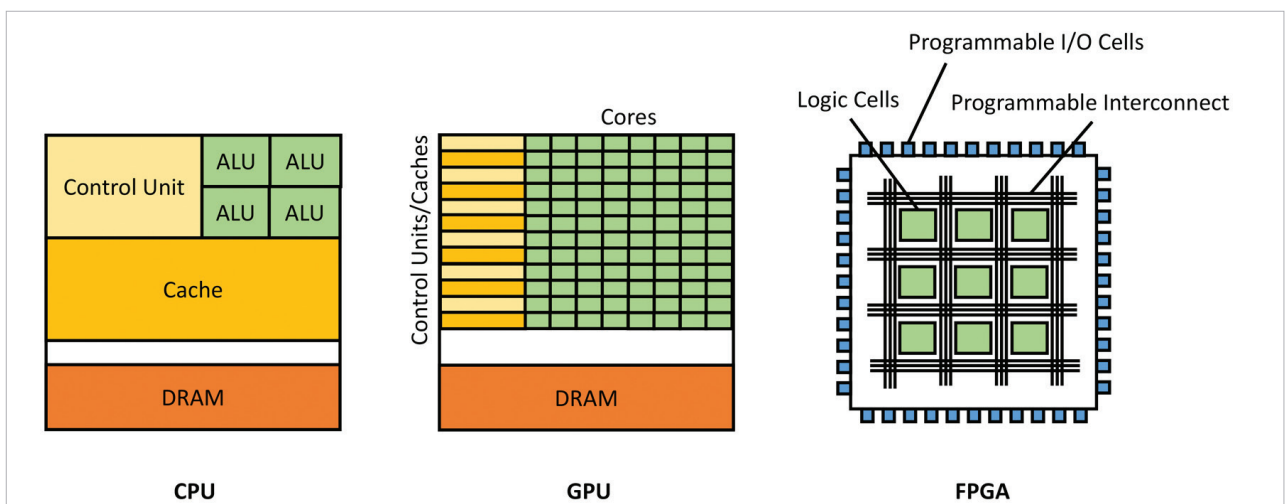


Figure 2: Different types of processors